THE SOLUTIONS

1. a) We would like to evaluate the double integral

$$I = \int_0^1 \left(\int_x^{\sqrt{2-x^2}} f(x,y)\, dy \right) dx\,.$$

Begin by sketching the region of integration, and rewrite the integral in terms of polar coordinates:

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta\,.$$

[You may assume the result $dx\,dy = r\,dr\,d\theta$.]

Now solve the integral for

$$f(x,y) = \sqrt{x^2+y^2}\left[\cos^{-1}\left(\frac{x}{\sqrt{x^2+y^2}}\right) + \sin^{-1}\left(\frac{y}{\sqrt{x^2+y^2}}\right)\right]\,.$$

[7]

Solution: The region is the one-eighths sector of the interior of the circle $x^2+y^2=2$, lying between the diagonal line $y=x$ and the $y-$axis. (Sketch at end.)
In polar coordinates, we have $\dfrac{\pi}{4} \le \theta \le \dfrac{\pi}{2}$ and $0 \le r \le \sqrt{2}$, so the integral will be

$$I = \int_{\theta=\pi/4}^{\pi/2} \left(\int_{r=0}^{\sqrt{2}} r\, f(r\cos\theta, r\sin\theta)\, dr \right) d\theta$$

Need to recognize that $f(x,y) =$

$$\sqrt{x^2+y^2}\left[\cos^{-1}\left(\frac{x}{\sqrt{x^2+y^2}}\right) + \sin^{-1}\left(\frac{y}{\sqrt{x^2+y^2}}\right)\right]$$
$$= r\left[\cos^{-1}\left(\frac{r\cos\theta}{r}\right) + \sin^{-1}\left(\frac{r\sin\theta}{r}\right)\right] = r(2\theta) \text{ so we have}$$

$$I = 2\left(\int_{\theta=\pi/4}^{\pi/2} \theta\, d\theta \right)\left(\int_{r=0}^{\sqrt{2}} r^2\, dr \right)$$

as the limits are constant and the integrand is a product. Finally

$$I = 2\left[\frac{1}{2}\theta^2\right]_{\pi/4}^{\pi/2}\ \left[\frac{1}{3}r^3\right]_0^{\sqrt{2}} = \frac{\pi^2\sqrt{2}}{8}\,.$$

Most errors related to not getting the area of integration right. Some did not recognize the simplification in $f(x,y)$ after converting to polar coordinates.

b) A function is defined as $f(x) = L - x$ on the interval $[0, L]$, where L is a positive constant. We would like to represent this function on $[0, L]$ as a Fourier cosine series.

(i) Let $g(x)$ be the appropriate extension of $f(x)$ so that the Fourier series for $g(x)$ will consist of a constant and cosine terms. Sketch $g(x)$ for $-2L < x < 2L$. [2]

(ii) Obtain the cosine series for $g(x)$ in terms of L, writing down the first five terms. [5]

(iii) If the first cosine term of the series is $\frac{4}{\pi}\cos x$, deduce the value of L and write out the series. [2]

(iv) Use Parseval's theorem and (i)-(iii) to find $\sum_{n=1}^{\infty} \frac{1}{n^4}$. [4]

Solution: (i) Need even extension, period $2L$. Sketch at end.

(ii) For the cosine series, we have $b_n = 0$ and

$$a_n = \frac{2}{2L}\int_{-L}^{L} g(x)\ \cos\left(\frac{2n\pi x}{2L}\right)\ dx\ =\ \frac{2}{L}\int_0^L (L-x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

using the half-range approach. For a_0 we integrate separately:

$$a_0\ =\ \frac{2}{L}\int_0^L L - x\, dx\ =\ L\,.$$

For the general term, we integrate by parts: $a_n =$

$$\frac{2}{L}\int_0^L (L-x)\cos\left(\frac{n\pi x}{L}\right)\ dx = \frac{2}{L}\left[(L-x)\frac{\sin(n\pi x/L)}{n\pi/L} + \frac{L}{n\pi}\int \sin(n\pi x/L)\right]_0^L$$

$$= \frac{2}{L}\left[0 - 0 - \frac{L}{n\pi}\frac{\cos(n\pi x/L)}{n\pi/L}\right]_0^L = -\frac{2L}{n^2\pi^2}(\cos(n\pi) - 1) = \begin{cases} 0 & \text{for even } n \\ \dfrac{4L}{n^2\pi^2} & \text{for odd } n \end{cases}$$

so the Fourier series is

$$g(x) = \frac{1}{2}L + \frac{4L}{\pi^2}\left[\frac{1}{1^2}\cos\left(\frac{\pi x}{L}\right) + \frac{1}{3^2}\cos\left(\frac{3\pi x}{L}\right) + \frac{1}{5^2}\cos\left(\frac{5\pi x}{L}\right) + \frac{1}{7^2}\cos\left(\frac{7\pi x}{L}\right) + \cdots\right]$$

(iii) Given that the first cosine term is $\frac{4}{\pi}\cos x$, we must have $L = \pi$, and the series is

$$g(x) = \frac{\pi}{2} + \frac{4}{\pi}\left[\cos x + \frac{1}{3^2}\cos 3x + \frac{1}{5^2}\cos 5x + \frac{1}{7^2}\cos 7x + \cdots\right]$$

(iv) Without knowledge of L, can use (ii), but messy! With $L=\pi$, we use Parseval's theorem for period 2π, and we have

$$\frac{2}{2\pi}\int_{-\pi}^{\pi} f^2(x)\,dx = \frac{1}{2}a_0^2 + \sum a_n^2$$

so we can integrate on $[0,\pi]$ and double the result:

$$\frac{2}{\pi}\int_0^{\pi}(\pi-x)^2\,dx \;=\; \frac{2\pi^2}{3} = \frac{1}{2}\pi^2 + \frac{16}{\pi^2}\sum_{\substack{n=1\\ n\ odd}}^{\infty}\frac{1}{n^4}$$

so that

$$\sum_{\substack{n=1\\ n\ odd}}^{\infty}\frac{1}{n^4} = \frac{\pi^4}{96}$$

Finally, let $m=2n$ for even integers, and write

$$S=\sum_{n=1}^{\infty}\frac{1}{n^4} = \sum_{\substack{n=1\\ n\ odd}}^{\infty}\frac{1}{n^4} + \sum_{\substack{n=1\\ n\ even}}^{\infty}\frac{1}{n^4} = \frac{\pi^4}{96} + \sum_{m=1}^{\infty}\frac{1}{(2m)^4} = \frac{\pi^4}{96} + \frac{1}{16}\sum_{m=1}^{\infty}\frac{1}{m^4} = \frac{\pi^4}{96} + \frac{1}{16}S$$

and $\left(1-\frac{1}{16}\right)S = \frac{\pi^4}{96} \Rightarrow S = \frac{\pi^4}{90}$.

A lot of partial answers and guesswork here. Many found the FS with L difficult, got something slightly workable and then tried to force it into the form required for (iii). Parseval's theorem applied with variable results, again lots of guesswork.

c) In your own words, explain:
(i) how to decide the existence of a function $F(x,y)$, so that a first-order Ordinary Differential Equation (ODE)

$$M(x,y)\frac{dy}{dx} + N(x,y) = 0$$

can be written in terms of the total differential dF, and
(ii) how a solution of the ODE is obtained by first finding $F(x,y)$. [5]

Solution: If $F(x,y)$ exists then

$$dF = \frac{\partial F}{\partial y}\,dy + \frac{\partial F}{\partial x}\,dx$$

and writing the ODE as

$$M(x,y)\,dy + N(x,y)dx = 0$$

and equating

$$M(x,y) = \frac{\partial F}{\partial y} \quad \text{and} \quad N(x,y) = \frac{\partial F}{\partial x}$$

we can test for the existence of F by checking

$$\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial M}{\partial x} = \frac{\partial N}{\partial y} = \frac{\partial^2 F}{\partial y \partial x}$$

Hence solve for F by integrating

$$F = \int M(x,y)\, dy + f(x) \quad \text{and} \quad F = \int N(x,y)\, dx + g(y)$$

where $f(x)$ and $g(y)$ are arbitrary functions, obtained by equating both expressions. The solution is then $dF = 0 \Rightarrow F =$constant.

Some very good answers, some very poor, and lots in-between. Many had clearly learned how to apply the algorithm, as evidenced by their answers to Q2a, but not understood its origin.

2. a) Solve the first-order ODE

$$\left(\frac{1}{2}t^3e^{\frac{1}{2}x} - t\cos x + 2x\right)\frac{dx}{dt} = t^2 + \sin x - 3t^2e^{\frac{1}{2}x},$$

subject to the initial condition $x(1) = 0$. [5]

Solution: Identify

$$M = \frac{1}{2}t^3e^{\frac{1}{2}x} - t\cos x + 2x, \quad \text{and} \quad N = -t^2 + 3t^2e^{\frac{1}{2}x} - \sin x\,.$$

Check that $\dfrac{\partial M}{\partial t} = \dfrac{\partial N}{\partial x}$. As

$$\frac{\partial M}{\partial t} = \frac{3}{2}t^2e^{\frac{1}{2}x} - \cos x = \frac{\partial N}{\partial x},$$

the condition is satisfied and the equation is exact, and there is a function $F(x,t)$ with $\dfrac{\partial F}{\partial x} = M$, and $\dfrac{\partial F}{\partial t} = N$. Hence

$F = \displaystyle\int M dx = t^3e^{\frac{1}{2}x} - t\sin x + x^2 + f(t)$, and

$F = \displaystyle\int N dt = t^3e^{\frac{1}{2}x} - t\sin x - \frac{1}{3}t^3 + g(x)$,

where $f(t)$ and $g(x)$ are arbitrary functions of t, resp. x. Comparing the two results for F, deduce that we need $f(t) = -\dfrac{1}{3}t^3$ and $g(x) = x^2$, giving the solution $F = C$ where C is a constant:

$$t^3e^{\frac{1}{2}x} - t\sin x - \frac{1}{3}t^3 + x^2 = C\,.$$

Finally implement the initial condition $x(1) = 0$ to find

$$1^3e^0 - 1\sin 0 - \frac{1}{3}1^3 + 0^2 = C \Rightarrow C = \frac{2}{3}$$

so the solution is

$$t^3e^{\frac{1}{2}x} - t\sin x - \frac{1}{3}t^3 + x^2 = \frac{2}{3}\,.$$

Mostly done well, a few trivial errors, but usually of signs. A minority was misled by having N on the RHS. Another minority found that $\dfrac{\partial M}{\partial x} \neq \dfrac{\partial N}{\partial t}$*, and were stuck.*

b) (i) Find and classify the stationary points of the function

$$f(x,y) = (y-x)(y^2-x)\,.$$ [5]

(ii) In your own words, explain why a saddle point lies at the intersection of two contour lines, and how this can be used to sketch the contours of a function of two variables. A sketch may be useful here! [5]

(iii) Sketch the contour where $f(x,y)=0$ and then construct the remaining contouirs of $f(x,y)$, carefully explaining how you use (i) and (ii). [5]

Solution: (i) Multiply out $f(x,y)=y^3-xy^2-xy+x^2$ and differentiate:

$$f_x=-y^2-y+2x=0 \quad \text{and} \quad f_y=3y^2-2xy-x=0$$

From the first equation, $2x=y^2+y$, and substitute this into the second equation:

$$3y^2-(y^2+y)y-\frac{1}{2}(y^2+y)=-y^3+\frac{3}{2}y^2-\frac{1}{2}y=-\frac{1}{2}y(y-1)(2y-1)=0$$

so $y=0,1,1/2$ giving stationary points $P_1(0,0)$, $P_2(1,1)$ and $P_3(3/8,1/2)$.

To classify, obtain the Hessian determinant

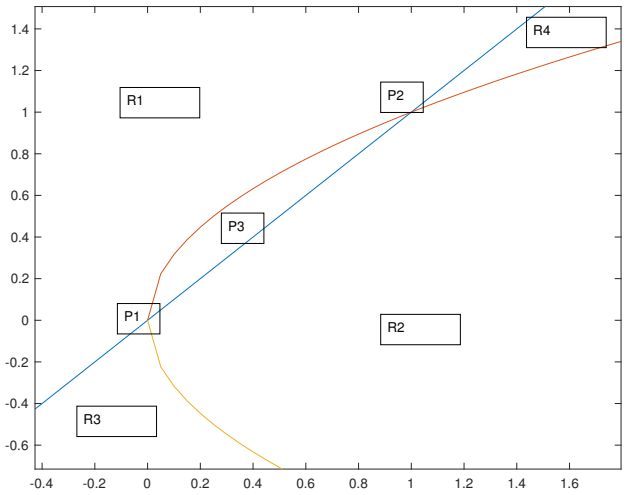
$$D=f_{xx}f_{yy}-(f_{xy})^2=2(6y-2x)-(-2y-1)^2$$

and $D(P_1)=D(0,0)=2(0-0)-(0-1)^2<0$ so $P_1(0,0)$ is a saddle point;
$D(P_2)=D(1,1)=2(6-2)-(-3)^2<0$ so $P_2(1,1)$ is also a saddle point.
Finally, $D(P_3)=D(3/8,1/2)=2(3-3/4)-(-2)^2>0$ and as $f_{xx}=2$, we have that $P_3(3/8,1/2)$ is a minimum.

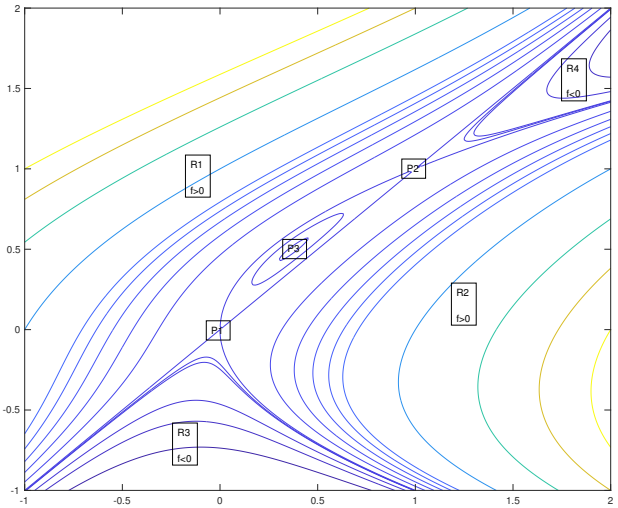
(ii) If we take $f(x,y)=C$ to be the value of $f(x,y)$ at the saddle, then moving away from the saddle in two directions, say North and South, without loss of generality, f will increase and moving in the other two directions, East and West, f will decrease. In between each increase and decrease, there is a direction in which f stays constant, so $f=C$, as we move (always wlog) towards the NE, SW, SE and NE. Thus, there are two contours intersecting at the saddle point: one moving from *NE* to *SW* and the other from *NW* to *SE*. A picture may help here, sketch at end.

Once the contours which meet at saddles are in place, the rest are easy as we know the value of f on the saddle contours and that, wherever we are, f will always increase on one side of the saddle contour and decrease on the other.

(iii) Given $f(x,y)=(y-x)(y^2-x)$ is factorized, we set $f=0\Rightarrow x=y$ or $x=y^2$, giving the zero-contours which intersect at the two saddle points $P_1(0,0)$ and $P_2(1,1)$. The minimum at $P_3(3/8,1/2)$ is enclosed by the parabola and the diagonal between the saddles:



This divides the plane into five regions. The region enclosing the minimum and four others, labelled R1-R4. Around the minimum, we have $f < 0$ as we move up in any direction to the zero-contour surrounding P_3. Moving from P_3 to the right, we cross the diagonal into R2, going from $f < 0$ to $f = 0$ to $f > 0$, so R2 has $f > 0$ and diverging to infinity as we move away from the origin. Similarly, moving to the left from P_3, crossing the parabola into R1, where $f > 0$. Moving from R1 into R3, crossing the diagonal in the third quadrant, we move from $f > 0$ to $f = 0$ to $f < 0$, giving the behaviour in R3, with $f \to -\infty$, likewise crossing into R4, either from R1 or R2. This completes the picture:



Part (i) almost everyone got. Part (ii) was less successful: theory generally weak. Many tried to explain but mixed up cause and effect. Many ignored (iii) and went straight to (iv). Many mislabelled (iv) or left out labels. A lot of sloppy plots where contour lines were drawn further apart away from stationary points, rather than clearly closer together.

c) Given the matrix

$$A = \begin{pmatrix} -2 & 1 & 1 \\ 2 & 1 & -2 \\ 3 & 1 & \gamma \end{pmatrix},$$

find the value of γ that makes the matrix singular. Taking $\gamma = -3$, use Gaussian elimination to find the inverse of A. [5]

Solution: Expanding by the first row, the determinant is $-2(\gamma+2)-1(2\gamma+6)+1(2-3) = -4\gamma-11$, so $\gamma = -11/4$ will make the determinant zero and the matrix singular.

Let $\gamma = -3$ and use Gauss-Jordan elimination on $(A : I)$ to get the inverse. To avoid fractions at the first step begin with

$$\left(\begin{array}{ccc|ccc} -2 & 1 & 1 & 1 & 0 & 0 \\ 2 & 1 & -2 & 0 & 1 & 0 \\ 3 & 1 & -3 & 0 & 0 & 1 \end{array}\right) \overset{\sim}{R_1+R_3} \left(\begin{array}{ccc|ccc} 1 & 2 & -2 & 1 & 0 & 1 \\ 2 & 1 & -2 & 0 & 1 & 0 \\ 3 & 1 & -3 & 0 & 0 & 1 \end{array}\right) \overset{\sim}{\substack{R_2-2R_1 \\ R_3-3R_1}} \left(\begin{array}{ccc|ccc} 1 & 2 & -2 & 1 & 0 & 1 \\ 0 & -3 & 2 & -2 & 1 & -2 \\ 0 & -5 & 3 & -3 & 0 & -2 \end{array}\right)$$

To still avoid fractions, first multiply R_2 by -2:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & -2 & 1 & 0 & 1 \\ 0 & 6 & -4 & 4 & -2 & 4 \\ 0 & -5 & 3 & -3 & 0 & -2 \end{array}\right) \overset{\sim}{R_2+R_3} \left(\begin{array}{ccc|ccc} 1 & 2 & -2 & 1 & 0 & 1 \\ 0 & 1 & -1 & 1 & -2 & 2 \\ 0 & -5 & 3 & -3 & 0 & -2 \end{array}\right) \overset{\sim}{\substack{R_1-2R_2 \\ R_3+5R_2}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 4 & -3 \\ 0 & 1 & -1 & 1 & -2 & 2 \\ 0 & 0 & -2 & 2 & -10 & 8 \end{array}\right)$$

$$\overset{\sim}{-\frac{1}{2}R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 4 & -3 \\ 0 & 1 & -1 & 1 & -2 & 2 \\ 0 & 0 & 1 & -1 & 5 & -4 \end{array}\right) \overset{\sim}{R_2+R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 4 & -3 \\ 0 & 1 & 0 & 0 & 3 & -2 \\ 0 & 0 & 1 & -1 & 5 & -4 \end{array}\right) \text{ so } A^{-1} = \begin{pmatrix} -1 & 4 & -3 \\ 0 & 3 & -2 \\ -1 & 5 & -4 \end{pmatrix}$$

Mostly done well. Arithmetic errors.

3. a) (i) Give a sketch of the function

$$f(t) = \begin{cases} 0 & , \quad 0 < t \leq a \ , \\ t-a & , \quad a < t \leq 2a \ , \\ a & , \quad 2a < t \ . \end{cases}$$

where $a > 0$. Show that the Laplace transform of $f(t)$ is
$\bar{f}(s) \ = \ \frac{1}{s^2}\left(e^{-as} - e^{-2as}\right)$. [5]

(ii) Given the second-order Ordinary Differential Equation (ODE)

$$\frac{d^2x}{dt^2} + x = \begin{cases} 0 & , \quad 0 < t \leq \pi \ , \\ t-\pi & , \quad \pi < t \leq 2\pi \ , \\ \pi & , \quad 2\pi < t \ . \end{cases}$$

which satisfies $x(0) = x'(0) = 0$, use (i) and Laplace Transforms to solve the ODE.

Give the solution in the form

$$x(t) = \begin{cases} x_1 & , \quad 0 < t \leq \pi \ , \\ x_2 & , \quad \pi < t \leq 2\pi \ , \\ x_3 & , \quad 2\pi < t \ , \end{cases}$$

simplifying as much as possible. Take care to explain each step. [10]

Solution:

(i) Sketch at end. Begin by writing $f(t)$ in terms of the Heaviside function:

$$f(t) = (t-a)H(t-a) - (t-a)H(t-2a) + aH(t-2a) \ .$$

We need terms in the form $f(t-\text{constant})H(t-\text{constant})$ to apply the second shift theorem, so we write

$$f(t) = (t-a)H(t-a) - (t-2a)H(t-2a)$$

The second shift theorem gives the transforms:

$$\mathscr{L}[(t-C)H(t-C)] = \mathscr{L}[t]e^{-Cs} = \frac{1}{s^2}e^{-Cs} \ :$$

$$\bar{f}(s) = \frac{1}{s^2}e^{-as} - \frac{1}{s^2}e^{-2as} \ ,$$

as required.

(ii) Take Laplace transforms of both sides of the ODE, using (i), and obtain

$$(s^2+1)\bar{x}(s) = \frac{1}{s^2}e^{-\pi s} - \frac{1}{s^2}e^{-2\pi s}$$

so that

$$\bar{x}(s) = \frac{1}{s^2(s^2+1)}e^{-\pi s} - \frac{1}{s^2(s^2+1)}e^{-2\pi s} \ .$$

Using partial fractions

$$\frac{1}{s^2(s^2+1)} = \frac{1}{s^2} - \frac{1}{s^2+1}$$

and so

$$\bar{x}(s) = \left(\frac{1}{s^2} - \frac{1}{s^2+1}\right) e^{-\pi s} - \left(\frac{1}{s^2} - \frac{1}{s^2+1}\right) e^{-2\pi s} .$$

Using the second shift theorem,

$$x(t) = \mathscr{L}^{-1}\left[\frac{1}{s^2} - \frac{1}{s^2+1}\right]_{t\to t-\pi} H(t-\pi) - \mathscr{L}^{-1}\left[\frac{1}{s^2} - \frac{1}{s^2+1}\right]_{t\to t-2\pi} H(t-2\pi)$$

where each term is obtained by taking the inverse Laplace transform of $1/s^2 - 1/(s^2+1)$, then shifting t by an appropriate amount and multiplying by an appropriate Heaviside function. In steps,

$$\mathscr{L}^{-1}\left[\frac{1}{s^2} - \frac{1}{s^2+1}\right] = t - \sin t ,$$

so that

$$\begin{aligned} x(t) &= [t-\sin t]_{t\to t-\pi} H(t-\pi) - [t-\sin t]_{t\to t-2\pi} H(t-2\pi) \\ &= [t-\pi-\sin(t-\pi)]H(t-\pi) - [t-2\pi-\sin(t-2\pi)]H(t-2\pi) . \end{aligned}$$

Given that

$$\sin(t-\pi) = -\sin t , \qquad \sin(t-2\pi) = \sin t ,$$

this simplifies to

$$x(t) = (t-\pi+\sin t)H(t-\pi) - (t-2\pi-\sin t)H(t-2\pi) .$$

The first term switches on at $t=\pi$, the second term switches on at $t = 2\pi$, so before that $x = 0$, and we can write

$$x(t) = \begin{cases} x_1 = 0 & , \quad 0 < t \le \pi , \\ x_2 = t-\pi+\sin t & , \quad \pi < t \le 2\pi , \\ x_3 = t-\pi+\sin t - (t-2\pi-\sin t) = \pi + 2\sin t & , \quad 2\pi < t , \end{cases}$$

Many tried (i) without the Heaviside function and second shift theorem, making it rather difficult, but a few ot these succeeded. A minority wrote the laplace transform incorrectly in (ii), or did not write it with $a = \pi$, leading to errors. Partial fractions mostly correct, but inverse transform of the result very variable. Getting the simplified result in the form x_1, x_2, x_3 was out of reach of many.

b) Given the matrix

$$A = \begin{pmatrix} 2 & -3 & -1 \\ -2 & 3 & 2 \\ 1 & -2 & 3 \end{pmatrix} ;$$

(i) Using two row operations, explain in words why A cannot be written as the product $A = LU$ where L is lower triangular and U upper triangular. Hence, identify the permutation matrix P needed so that PA can be written as LU. [5]

(ii) Use Gaussian elimination to transform PA into the upper triangular U and write down the elimination matrices needed for this. Write down the inverses of the elimination matrices and find L, so that $PA = LU$. [5]

Solution: (i) The row operations needed produce $A_{21} = A_{31} = 0$:

$$A = \begin{pmatrix} 2 & -3 & -1 \\ -2 & 3 & 2 \\ 1 & -2 & 3 \end{pmatrix} \underset{\substack{R_2+R_1 \\ R_3-\frac{1}{2}R_1}}{\sim} \begin{pmatrix} 2 & -3 & -1 \\ 0 & 0 & 1 \\ 0 & -\frac{1}{2} & \frac{7}{2} \end{pmatrix}$$

The next operation to make A upper-triangular would be a permutation, swapping rows 2 and 3. Taken together we would have

$$MA = U$$

where M is the product of the elimination and permutation matrices. So $A = LU$ where L is the inverse of M. As M includes a permutation, its inverse will do as well, so L cannot be lower triangular. The $LU-$decomposition is only possible with elimination steps, not permutations. Hence we need

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

(ii) Begin with

$$PA = \begin{pmatrix} 2 & -3 & -1 \\ 1 & -2 & 3 \\ -2 & 3 & 2 \end{pmatrix} \underset{\substack{R_2-\frac{1}{2}R1 \\ R_3+R1}}{\sim} \begin{pmatrix} 2 & -3 & -1 \\ 0 & -\frac{3}{2} & \frac{7}{2} \\ 0 & 0 & 1 \end{pmatrix} = U$$

The elimination matrices corresponding to the two row operations and their inverses are

$$E_{21} = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, E_{21}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad E_{31} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, E_{31}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}.$$

Hence $L = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$.

Part (i) was done correctly by few, partially by many. Row operations were fine, but why a permutation would make LU impossible proved elusive. Part (ii) mostly fine, but a many didn't give the correct elimination matrices and inverses, even though they had LU correct.

4. a) You are given four points $A(-1,1,5)$, $B(-3,1,1)$, $C(1,1,0)$ and $D(2,-1,-2)$, with two lines, L_1 through A and B, and L_2 through C and D.

(i) Find the vector equations of L_1 and L_2, in terms of four vectors and two parameters, $\lambda, \mu \in \mathbb{R}$. Explain in your own words how to find points P_1 on L_1 and P_2 on L_2, so that $|\overrightarrow{P_1P_2}|$ is the shortest distance between the two lines. [5]

(ii) Write down the three equations in λ, μ and one further unknown, which you need to find P_1 and P_2. Write these equations in matrix form $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$. Is A invertible or singular, and how do we know this without any calculations? [4]

(iii) Use Gaussian elimination to obtain the echelon form of the augmented coefficient matrix $(A : \underline{\mathbf{b}})$, and hence find $\underline{\mathbf{x}}$. Hence, obtain P_1, P_2 and the shortest distance. Finally, consider the situation where this question is badly posed: the lines intersect and the minimum distance is zero. Explain how A would change in this case. [4]

Solution:

(i) Use $\overrightarrow{AB} = (-3,1,1) - (-1,1,5) = (-2,0,-4)$ as the direction vector for L_1 which then has equation

$$\underline{\mathbf{r}} = (-1,1,5) + \lambda(-2,0,-4) = \underline{\mathbf{p}} + \lambda\underline{\mathbf{a}},$$

where $\lambda \in \mathbb{R}$ Likewise L_2 has direction vector $\overrightarrow{CD} = (2,-1,-2) - (1,1,0) = (1,-2,-2)$ and equation

$$\underline{\mathbf{r}} = (2,-1,-2) + \mu(1,-2,-2) = \underline{\mathbf{q}} + \mu\underline{\mathbf{b}},$$

with $\mu \in \mathbb{R}$.

A shortest distance is always perpendicular, so the vector $\underline{\mathbf{x}} = \overrightarrow{P_1P_2}$ is perpendicular to both lines, and its direction is given by the vector product of the direction vectors, $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$, of the two lines. Therefore $\underline{\mathbf{x}} = k(\underline{\mathbf{a}} \times \underline{\mathbf{b}})$ where $k \in \mathbb{R}$. We can also write P_1 in terms of its coordinates on L_1, given as $\underline{\mathbf{p}} + \lambda\underline{\mathbf{a}}$, similarly P_2 as $\underline{\mathbf{q}} + \mu\underline{\mathbf{b}}$. Subtracting we have two expressions:

$$\overrightarrow{P_1P_2} \;=\; \underline{\mathbf{q}} + \mu\underline{\mathbf{b}} - (\underline{\mathbf{p}} + \lambda\underline{\mathbf{a}}) = k\underline{\mathbf{a}} \times \underline{\mathbf{b}}\,.$$

This is a system of three equations in three unknowns, μ, λ and k. Solve and substitute back to find P_1 and P_2.

(ii) Following from (i), we require the vector product

$$\underline{\mathbf{a}} \times \underline{\mathbf{b}} = (-2,0,-4) \times (1,-2,-2) \;=\; (-8,-8,4)$$

so take $\underline{\mathbf{x}} = k(2,2,-1)$, for simplicity. Hence

$$\overrightarrow{P_1P_2} = \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix} - \left[\begin{pmatrix} -1 \\ 1 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 0 \\ -4 \end{pmatrix} \right] = k \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$$

which represents three equations in three unknowns:

$$\begin{aligned} \mu + 2\lambda - 2k &= -3\,, \\ -2\mu \qquad - 2k &= 2\,, \\ -2\mu + 4\lambda + k &= 7\,. \end{aligned}$$

In matrix form we have

$$A\underline{\mathbf{x}} = \begin{pmatrix} 1 & 2 & -2 \\ -2 & 0 & -2 \\ -2 & 4 & 1 \end{pmatrix} \begin{pmatrix} \mu \\ \lambda \\ k \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \\ 7 \end{pmatrix} = \underline{\mathbf{b}}\,.$$

Finally, A must be invertible, as we expect a unique solution μ, λ, k.

(iii) Using row operations on the augmented matrix $(A : \underline{\mathbf{b}})$ we have

$$\begin{pmatrix} 1 & 2 & -2 & | & -3 \\ -2 & 0 & -2 & | & 2 \\ -2 & 4 & 1 & | & 7 \end{pmatrix} \underset{R3+2R_1}{\overset{R_2+2R_1}{\sim}} \begin{pmatrix} 1 & 2 & -2 & | & -3 \\ 0 & 4 & -6 & | & -4 \\ 0 & 8 & -3 & | & 1 \end{pmatrix} \overset{}{\underset{R_3-2R_2}{\sim}} \begin{pmatrix} 1 & 2 & -2 & | & -3 \\ 0 & 4 & -6 & | & -4 \\ 0 & 0 & 9 & | & 9 \end{pmatrix}$$

This is echelon form and the last row gives $k = 1$. Backsubstitution: row 2 has $4\lambda - 6k = -4$ so with $k = 1$ we have $\lambda = 1/2$. Finally, row 1 has $\mu + 2\lambda - 2k = -3$ and substituting k, λ we obtain $\mu = -2$, and we have $\underline{\mathbf{x}} = (-2, 1/2, 1)$.

Substituting $\mu = -2$ into the equation for L_2 we get $P_2(0,3,2)$; likewise $\lambda = 1/2$ into L_1 we get $P_1(-2,1,3)$. This confirms that

$$\overrightarrow{P_1P_2} = (0,3,2) - (-2,1,3) = (2,2,-1) = k(2,2,-1)$$

and the distance is $|\overrightarrow{P_1P_2}| = \sqrt{4+4+1} = 3$.

If the lines intersect, the distance is zero, so $P_1 = P_2$ and $\overrightarrow{P_1P_2}$ is the zero vector. However, nothing else would change: A would be the same, and the solution would find $k = 0$

Almost everyone got the equations in (i) but found the explanation difficult. Similarly, writing the matrix in (ii) was easy, but not the idea of being able to identify the matrix as invertible and why. Many solved it correctly but omitted detail in (iii).

b) Use Gaussian elimination to find the reduced echelon form of

$$A = \begin{pmatrix} 3 & 4 & 2 & -1 \\ 2 & 6 & -7 & 11 \\ 1 & 2 & -1 & 2 \end{pmatrix}.$$

Identify rank, pivot columns and free columns. Obtain bases for the four fundamental subspaces of A: column space, row space, nullspace and left nullspace, taking to care to explain how you obtain these. Confirm the relationships between the dimensions of the subspaces and the number of rows and columns of A. [12]

Solution: Row operations give the reduced echelon form: $A \sim$

$$R_1 \leftrightarrow R_3 \begin{pmatrix} 1 & 2 & -1 & 2 \\ 2 & 6 & -7 & 11 \\ 3 & 4 & 2 & -1 \end{pmatrix} \sim \begin{matrix} R_2 - 2R_1 \\ R_3 - 3R_1 \end{matrix} \begin{pmatrix} 1 & 2 & -1 & 2 \\ 0 & 2 & -5 & 7 \\ 0 & -2 & 5 & -7 \end{pmatrix}$$

$$\begin{matrix} \sim \\ R_3+R_2 \\ \frac{1}{2}R_2 \end{matrix} \begin{pmatrix} 1 & 2 & -1 & 2 \\ 0 & 1 & -5/2 & 7/2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{matrix} \sim \\ R_1 - 2R_2 \end{matrix} \begin{pmatrix} \boxed{1} & 0 & 4 & -5 \\ 0 & \boxed{1} & -5/2 & 7/2 \\ 0 & 0 & 0 & 0 \end{pmatrix} = R$$

is the reduced echelon form, with pivots as marked. The first two columns are pivot columns, the second two are free columns, and the rank is 2.

For bases, the row space is the simplest one: we take linear combinations of the rows containing pivots, so a basis for $C(A^T)$ is

$$\begin{pmatrix} 1 \\ 0 \\ 4 \\ -5 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ 1 \\ -5/2 \\ 7/2 \end{pmatrix} .$$

For the column space, we identify the pivot columns, C_1 and C_2 and take as basis these two columns, not from R, but from A, so a basis for $C(A)$ is

$$\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 4 \\ 6 \\ 2 \end{pmatrix} .$$

The basis for the nullspace is obtained from the special solutions to $A\underline{\mathbf{x}} = \underline{\mathbf{0}}$, setting the free variables to $0, 1$ in the equations from R:

from row 1: $x_1 + 4x_3 - 5x_4 = 0$,
from row 2: $x_2 - \frac{5}{2}x_3 + \frac{7}{2}x_4 = 0$.
Letting $x_3 = 1$ and $x_4 = 0$ we get $x_2 = 5/2$ and $x_1 = -4$, so

$$\underline{\mathbf{x}}_1^s = \begin{pmatrix} -4 \\ 5/2 \\ 1 \\ 0 \end{pmatrix} .$$

Letting $x_3 = 0$ and $x_4 = 1$ we get $x_2 = -7/2$ and $x_1 = 5$ so

$$\underline{\mathbf{x}}_2^s = \begin{pmatrix} 5 \\ -7/2 \\ 0 \\ 1 \end{pmatrix} .$$

The basis for the nullspace $N(A)$, is $\underline{\mathbf{x}}_1^s$ and $\underline{\mathbf{x}}_2^s$.

Finally, for the left nullspace we require $\underline{\mathbf{x}}$ satisfying $A^T\underline{\mathbf{x}} = \underline{\mathbf{0}} \Leftrightarrow \underline{\mathbf{x}}^T A = (0,0,0,0)$. We note there is one row of zeroes in R. Applying the row operations that changed $R_3 \to (0,0,0,0)$ to the identity matrix, we get $I_3 \sim$

$$R_1 \leftrightarrow R_3 \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \sim \begin{matrix} R_2 - 2R_1 \\ R_3 - 3R_1 \end{matrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & -2 \\ 1 & 0 & -3 \end{pmatrix} \begin{matrix} \sim \\ R_3 + R_2 \end{matrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & -2 \\ 1 & 1 & -5 \end{pmatrix}$$

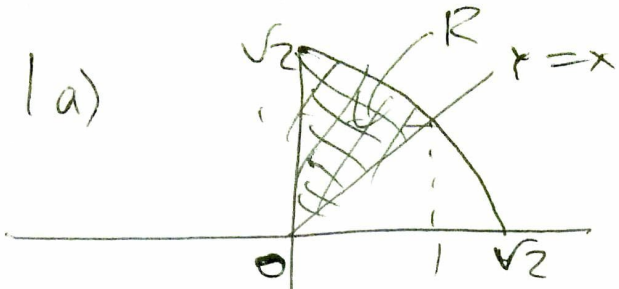
Multiplying A on the right by the last row of this matrix gave the row of zeroes in R, so $(1,1,-5)^T$ is a basis for the left nullspace $N(A^T)$.

As A is 3×4, we have
$m = 3 = 1+2 = \dim N(A^T)+\dim C(A)$, and $n = 4 = 2+2 = \dim N(A)+\dim C(A^T)$.

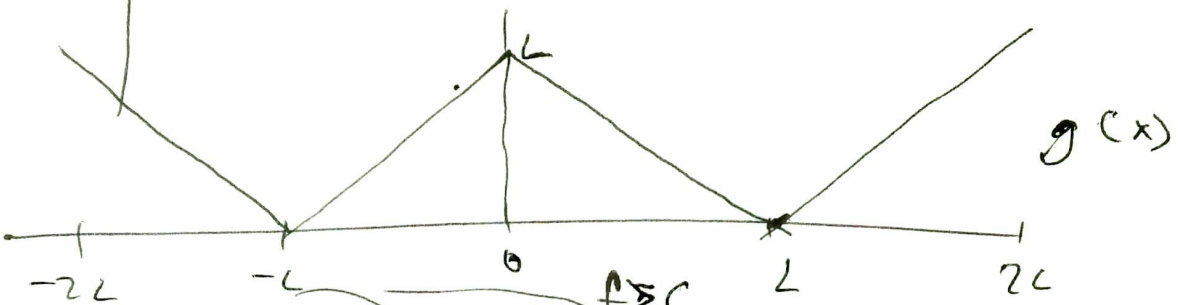
Poorly done, only a minority of students had absorbed this. Typical errors involved giving the basis of the column space as all four columns, or solving inhomogenous equations for the nullspace. Only a handful had understood the algorithm for finding the left nullspace, and most did not attempt this.

Sketches ELEC 40018 SOS 2021

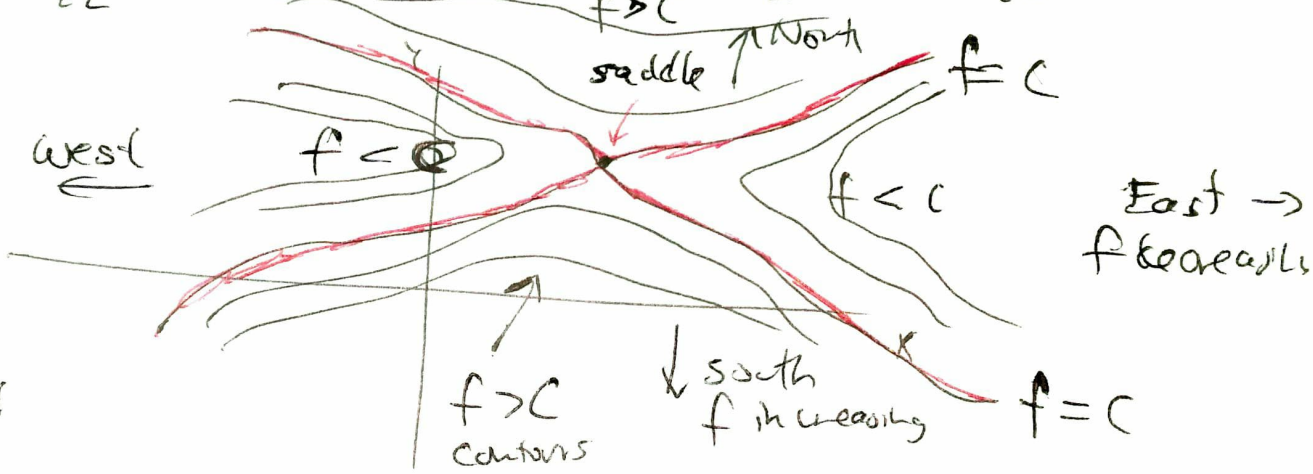
1a)



1b) (i)



2b)(ii)



f=C : red

3a)

